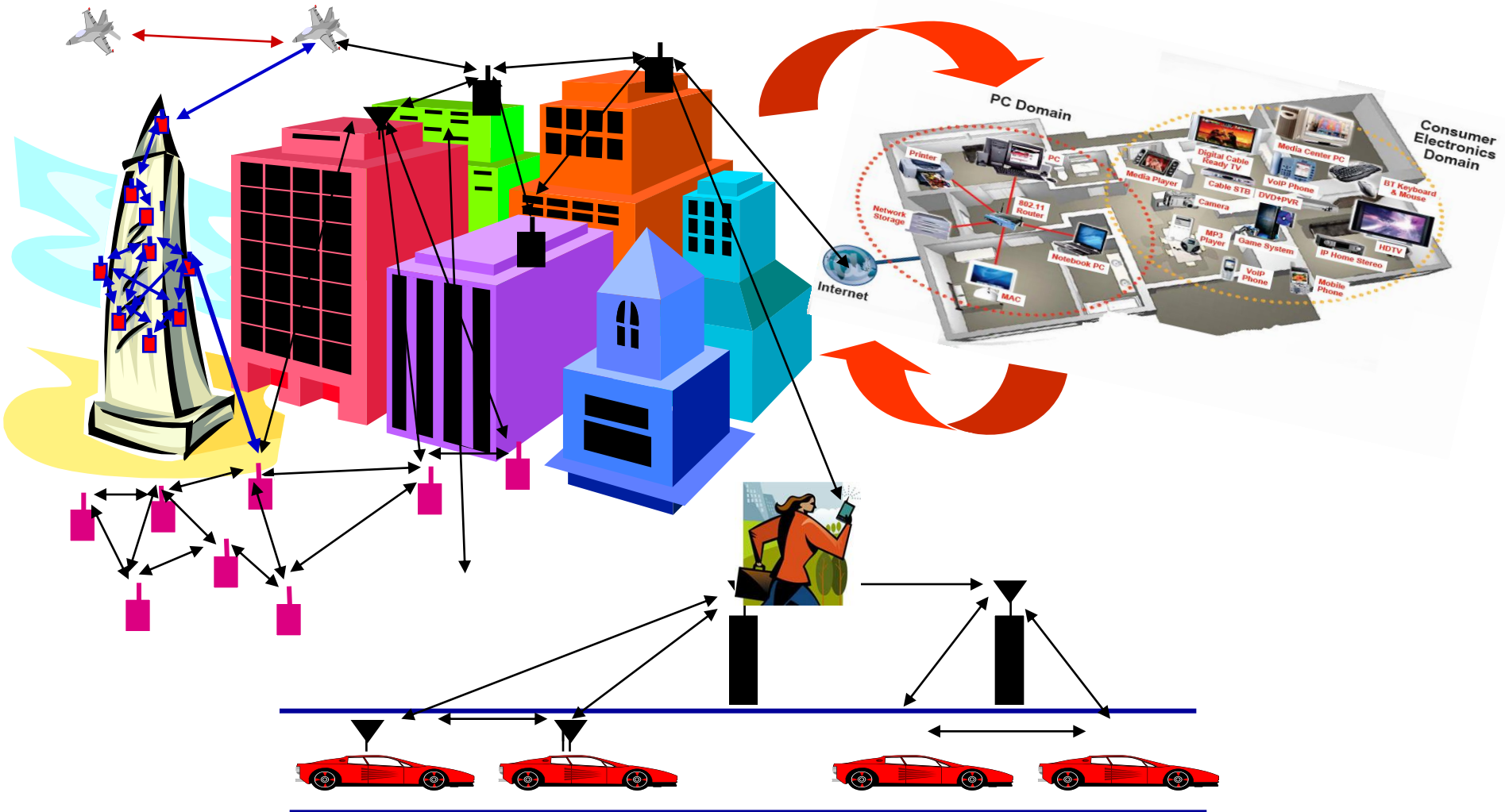


Wireless Digital Communications



《数字通信》

课程特点与学习方法

- 《数字通信》是本科《通信原理》课程的延续与深化，理论性较强。
- 《数字通信》是信息与通信工程专业研究生最重要的专业基础课程之一，就像《通信原理》是本科通信工程专业最重要的基础课程之一。
- 对于授课内容，要善于思考，对于课后习题，须勤学多练，有所感悟。
- 对大部分同学来说，《数字通信》是在课堂上系统学习通信基础理论的“最后一顿晚餐”。要珍惜机会，学有所获。

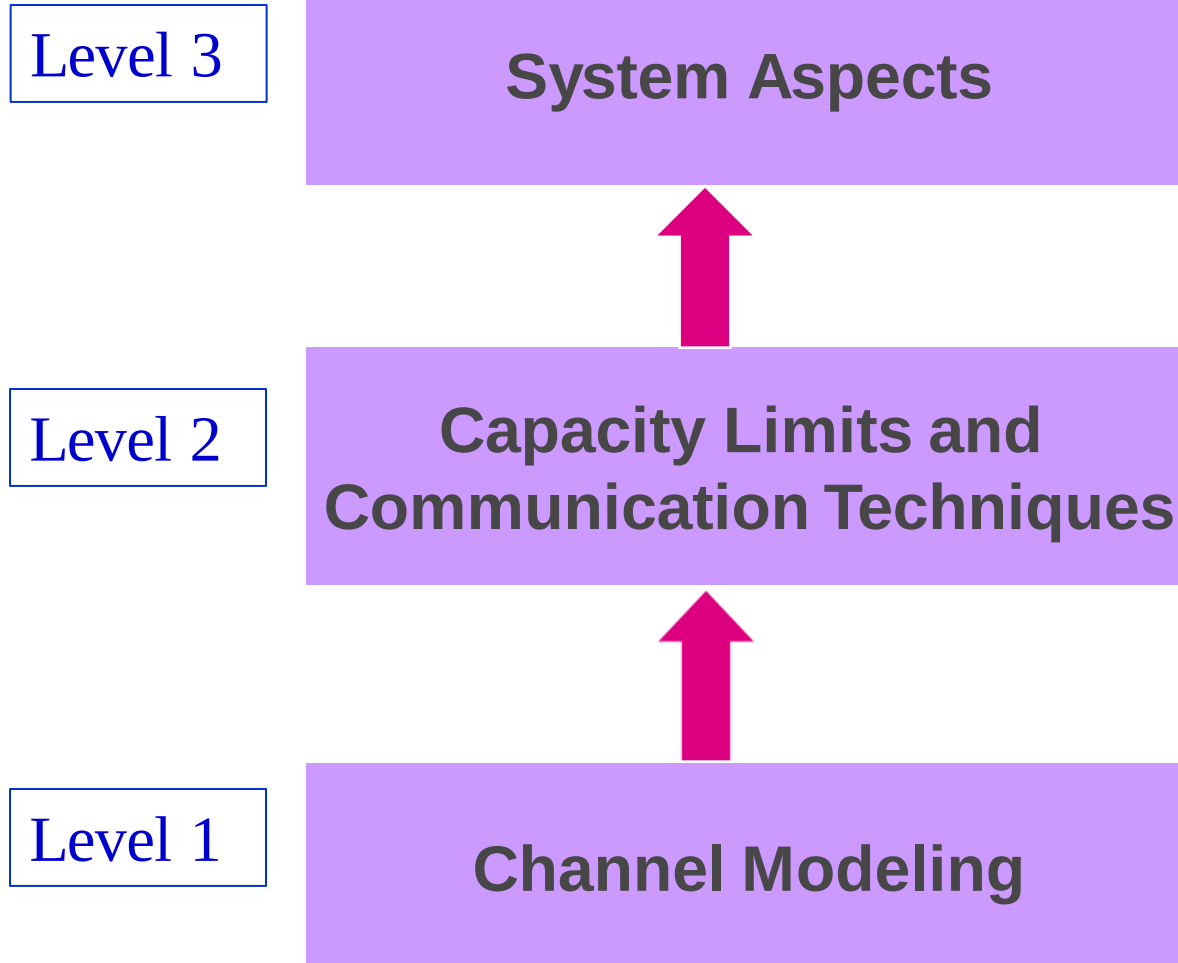
《数字通信》成绩评定

- 考试成绩占70% ， 平时成绩占30%
- 平时成绩包含：课后作业+签到

Course Information

- Prerequisites: Signal and Systems, Probability and Random Process, Communications Systems
- Required Textbook:
 - Wireless Communications* (by Andrea Goldsmith)
 - Fundamentals of Wireless Communications*
(by David Tse etc.)
 - Principles of Mobile Communications, 2nd ed*
(by G. L. Stuber)
 - Digital Communications* (by J. Proakis)

Course Structure



Course Syllabus

- Overview of Wireless Communications
- Path Loss, Shadowing, and Fading Models (Level 1)
- Capacity of Wireless Channels (Level 2)
- Digital Modulation and its Performance (Level 2)
- Adaptive Modulation (Level 2)
- Diversity (Level 2)
- MIMO Systems (Level 2)
- Multicarrier Modulation (Level 2)
- Spread Spectrum (Level 2)
- Multiuser Communications & Wireless Networks (Level 3)

无线通信系统的常用性能指标

- 可靠性：信号干扰噪声比(SINR), 误码率 (BER)
- 有效性：频谱效率，吞吐量，支持连接数，能量效率等
- 即时性：系统传输与处理时延
- 复杂性：计算量

数据速率、时延和可靠性的相互折衷

A Shannon Theory View for SISO Channel

by	Achievable Data Rate: R	Latency (block-length: n)	Reliability (BER: P_e)	Remarks
Shannon, 1949	$R \rightarrow C = B \cdot \log_2(1 + \text{SNR})$	$n \rightarrow \infty$	$P_e \rightarrow 0$	C : Shannon Capacity
Gallager, 1956	$R < C$	for given n	$P_e < 2^{-nE_r(R)}$	Gallager exponent: $E_r(R) > 0, R < C$
Shannon, 1959	$R \approx C - \frac{\Phi^{-1}(P_e)}{\sqrt{n}} \sqrt{\frac{2 + \text{SNR}}{2 \cdot \text{SNR}(1 + \text{SNR})}}$	for given n	for given P_e	Φ^{-1} : the reverse Gaussian cumulant function

MIMO 系统

速率与可靠性的折衷

Taking a MIMO system with $N_t \times N_r$ antennas as an example under the *i. i. d.* assumption.

When a spatial multiplex gain $r < \min. \{N_t, N_r\}$ is chosen, the **achievable data rate** will be

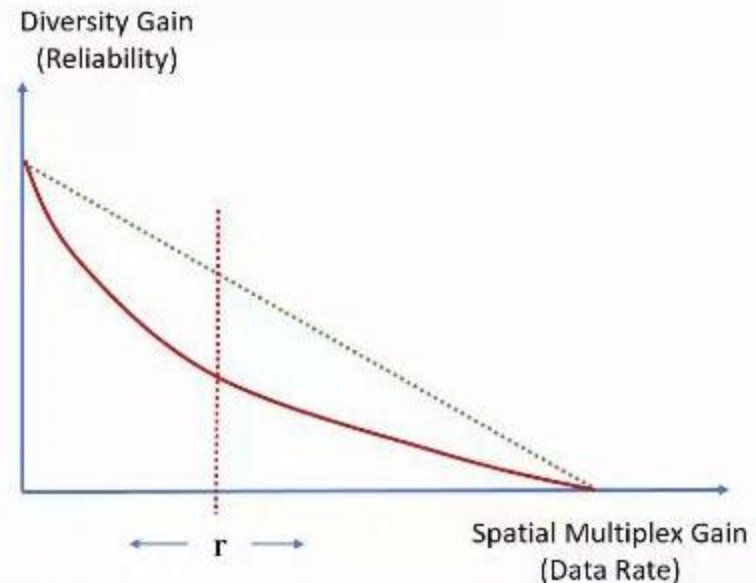
$$R = r \log_2(1 + \text{SNR})$$

and the **diversity gain** will be*

$$d = (N_t - r + 1)(N_r - r + 1)$$

which implies the following link **reliability**

$$P_e \propto \text{SNR}^{-(N_t - r + 1)(N_r - r + 1)}.$$

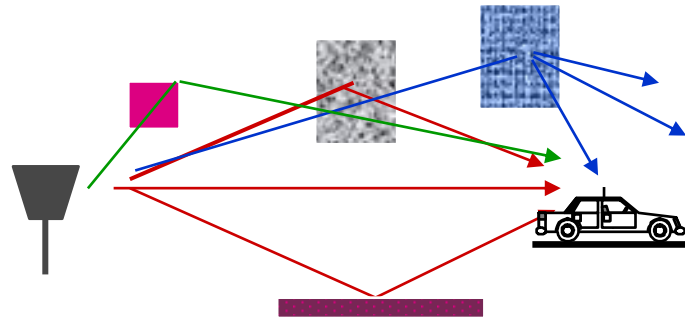


* L. Zheng, and D. N. C. Tse, "Diversity and multiplexing: A fundamental tradeoff in multiple-antenna channels," *IEEE Transactions on Information Theory*, Vol.49, no.5, pp.1073-1096, May 2003

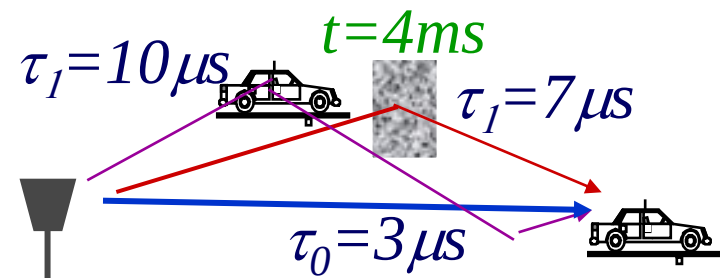
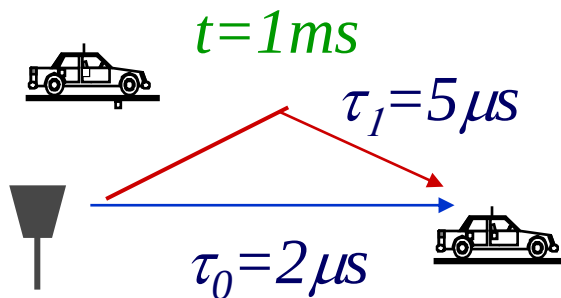
无线信道的第一类特征

——多径性和时变性

- 多径性

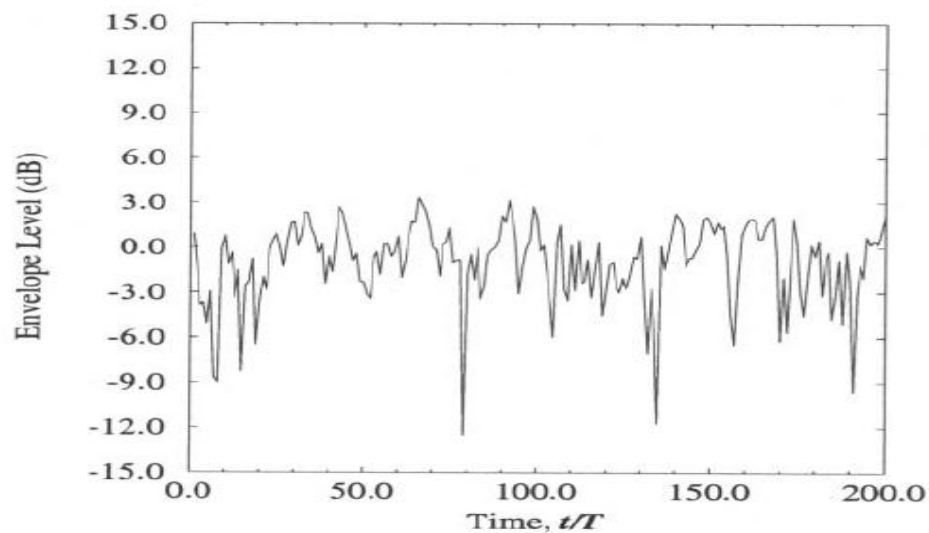


- 时变性



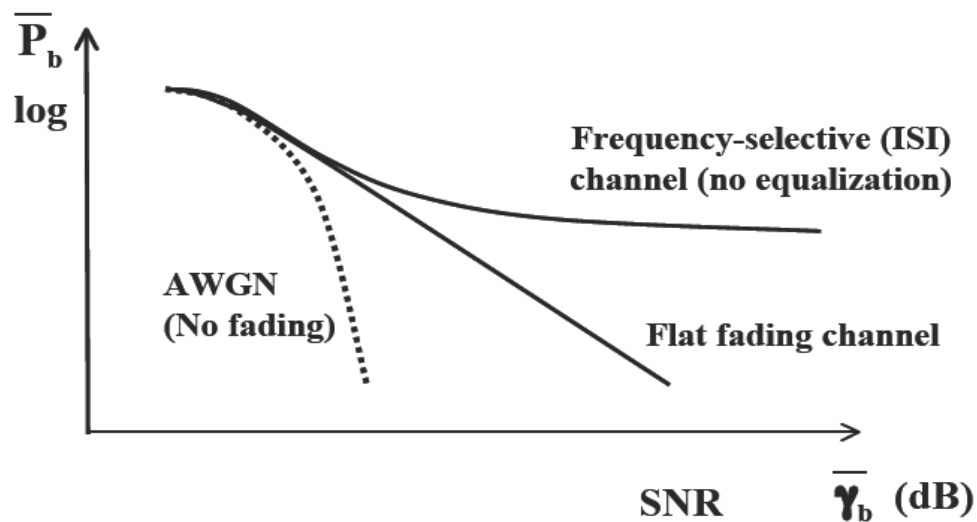
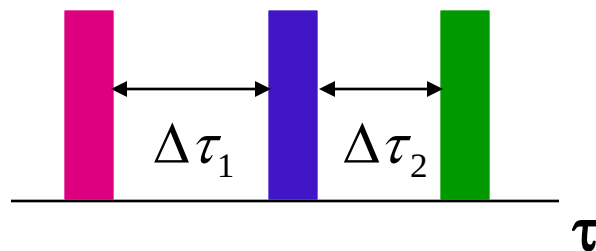
时变性和多径性对窄带传输的影响

——时变衰落



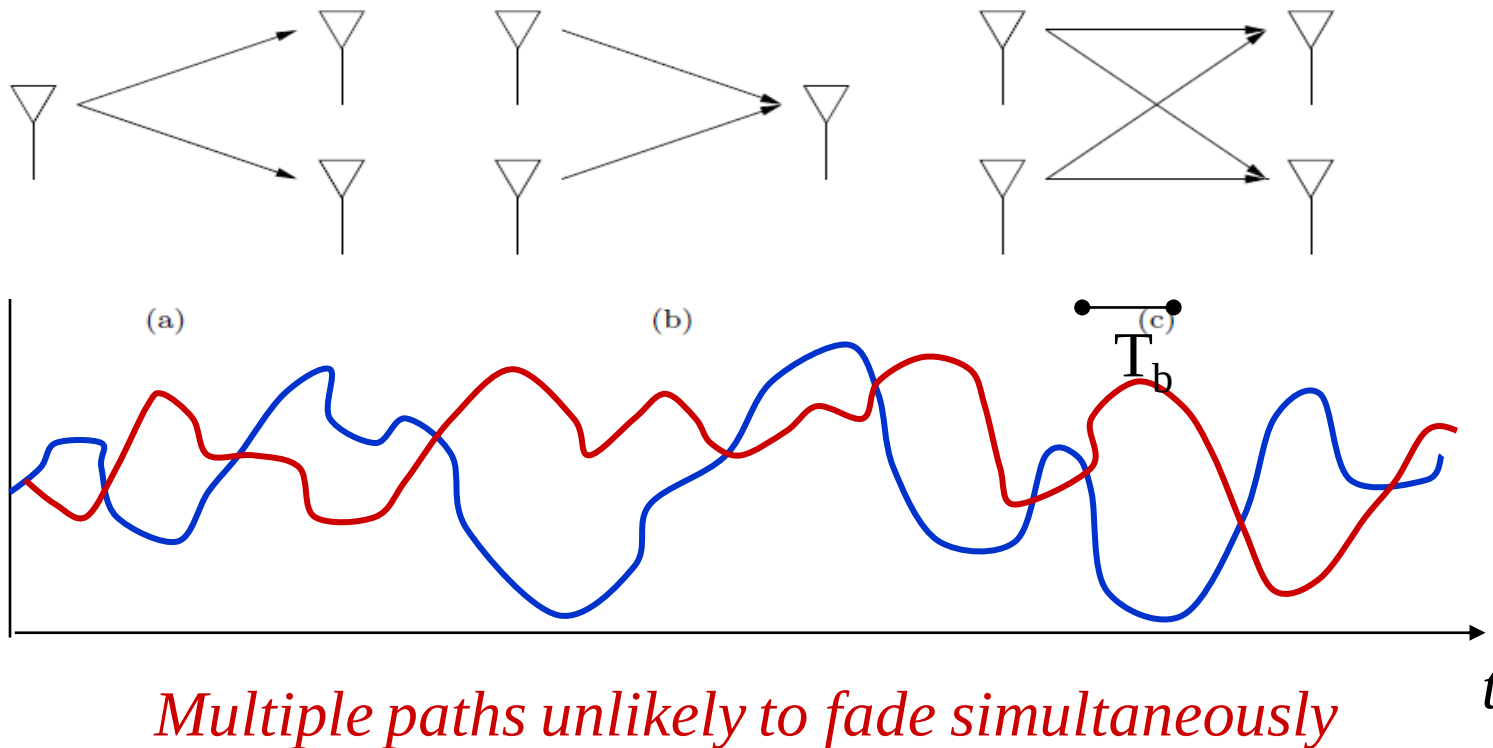
时变性和多径性对宽带传输的影响

——衰落和符号间干扰

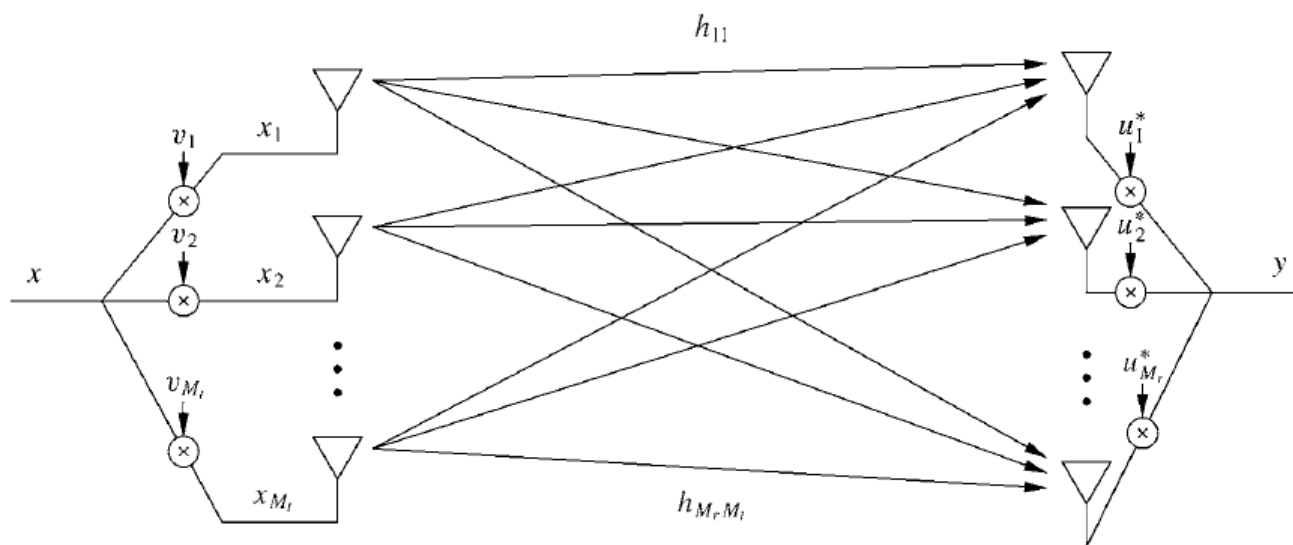


改善可靠性(抗衰落)的有效措施 ——空间分集

- Receive Diversity, Transmit Diversity, and Receive-Transmit Diversity



空间分集的一般形式



改善可靠性(抗符号间干扰)的有效措施

——多载波传输

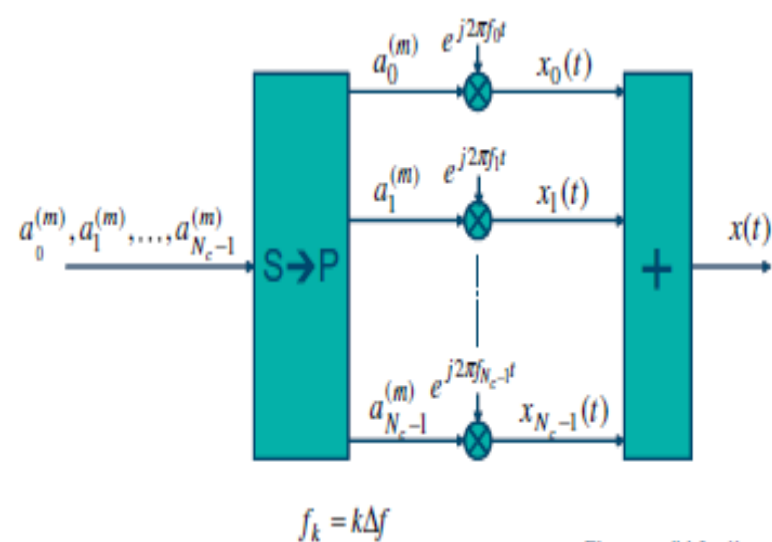
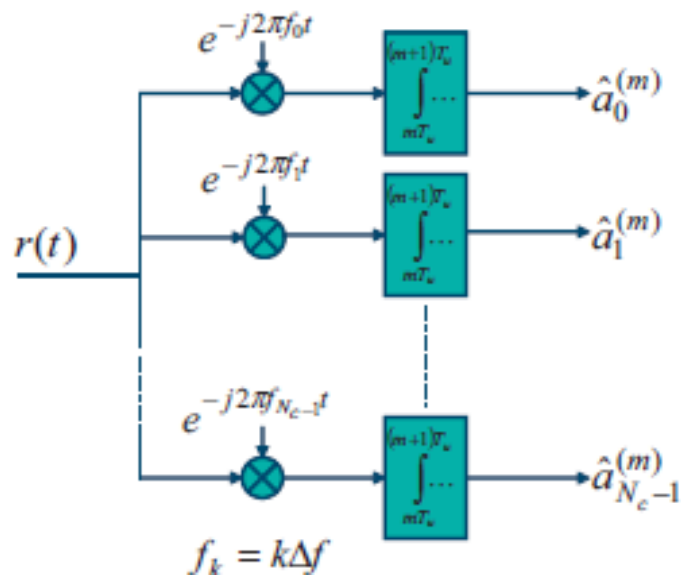


Figure valid for time interval $mT_u \leq t < (m+1)T_u$



提高频谱效率的有效措施

——空间复用

MIMO Channel Capacity (Telatar, 1999):

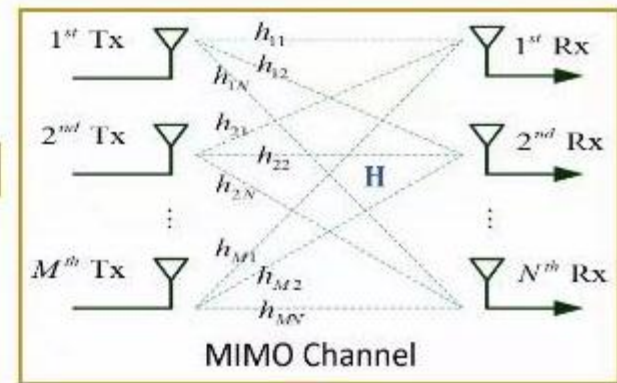
$$\mathcal{C} = \log_2 \left(\det \left(1 + \frac{1}{\sigma^2} \mathbf{H} \mathbf{Q} \mathbf{H}^\dagger \right) \right)$$

Its optimal solution:

$$\max_{\mathbf{Q}} \mathcal{C} = \sum_i (\log_2(\mu \lambda_i))^+ \quad \text{s.t.} \quad \sum_i \left(\mu - \frac{1}{\lambda_i} \right)^+ = P$$

λ_i : the eigen values of $\mathbf{H} \mathbf{H}^\dagger$

P : total Tx power σ^2 : noise power



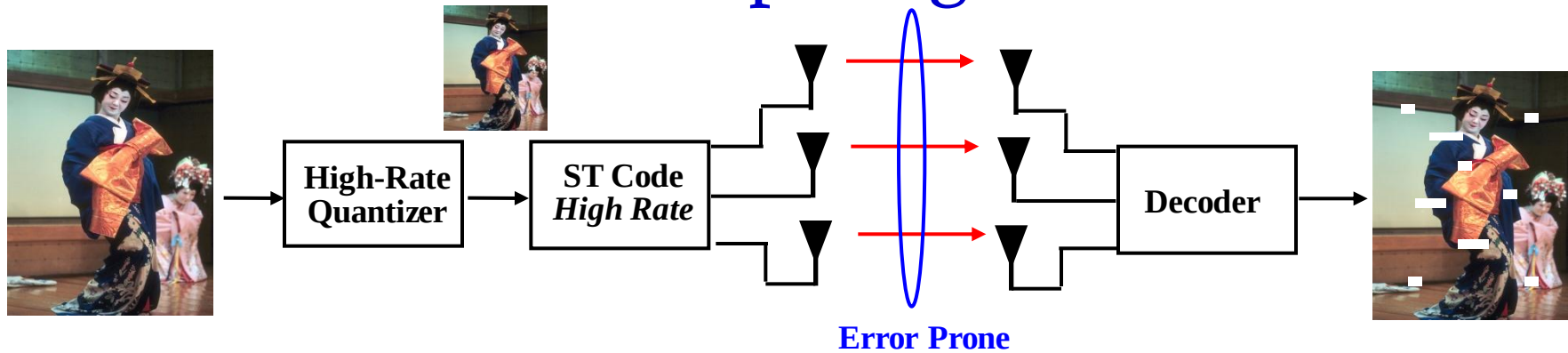
In the special *i. i. d.* case:

$$\mathcal{C} \rightarrow \min\{M, N\} \times \log_2(1 + \text{SNR})$$

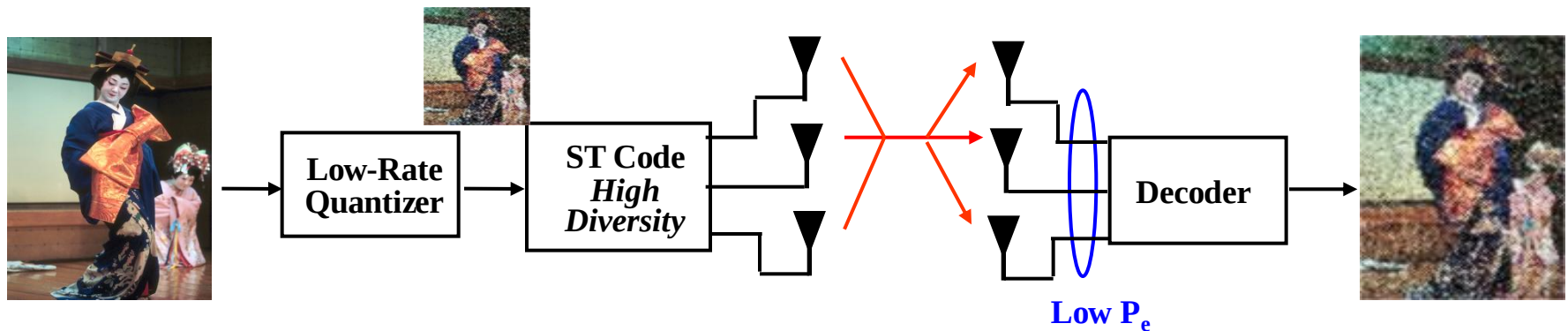
Spatial Multiplex Gain !!!

分集与复用的比较

- Use antennas for multiplexing:

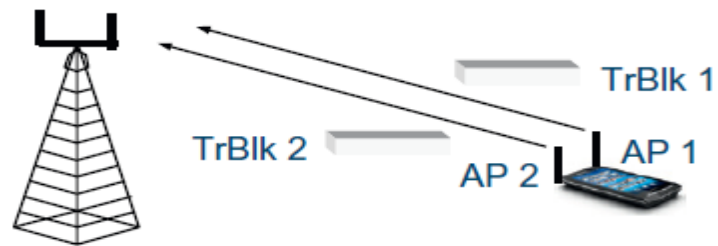


- Use antennas for diversity

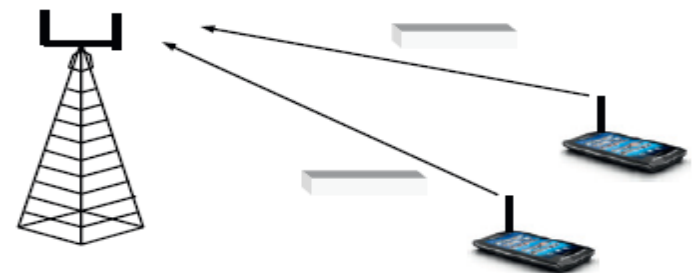


Depends on end-to-end metric: *Solve by optimizing app. metric*

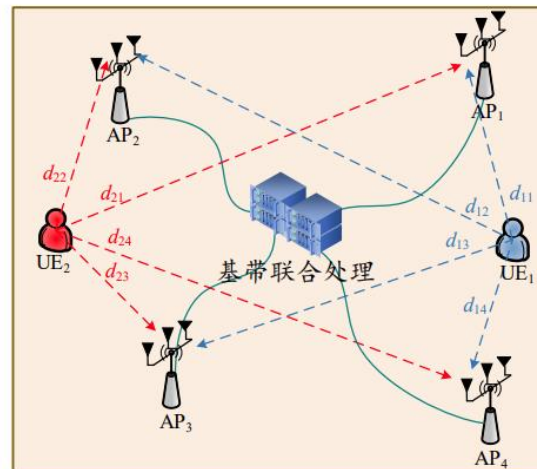
从SU-MIMO到MU-MIMO 再到Cell-Free MIMO



SU-MIMO



MU-MIMO

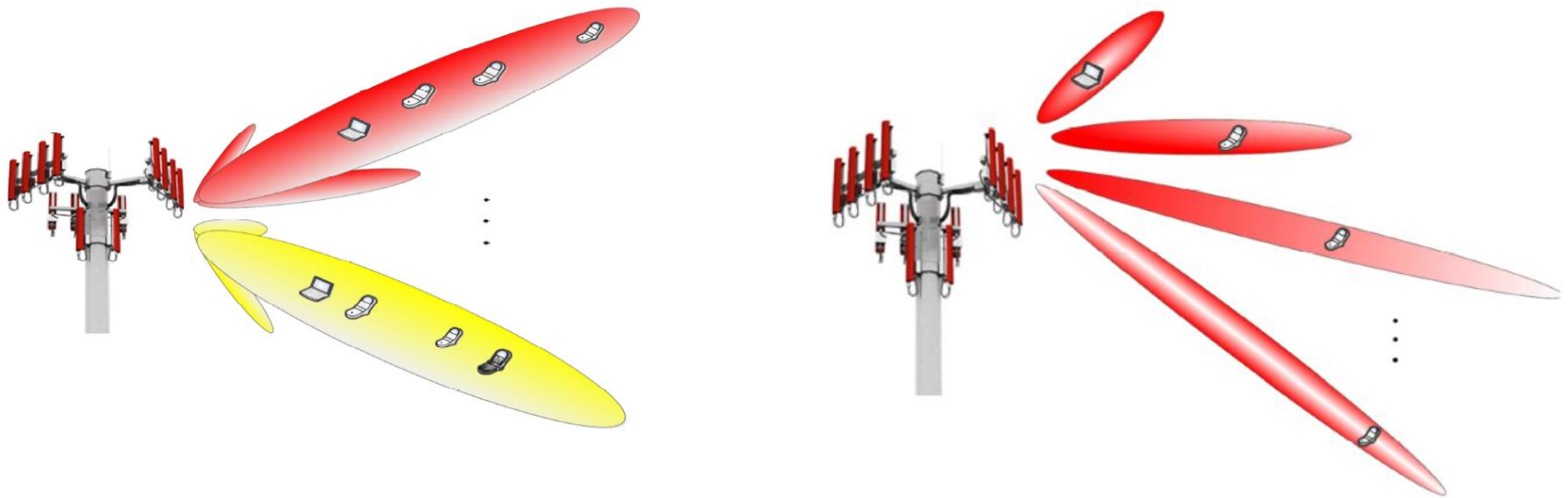


Cell-Free based MP-to-MP MIMO

从SU-MIMO到MU-MIMO 再到Cell-Free MIMO

- **Point-to-point MIMO (SU-MIMO):** 😞
Single user MIMO: the SE is usually limited by the maximally implementable antenna number at UE.
- **Point to multi-point MIMO (MU-MIMO):** 😊
Single BS serves multiple UEs by using a same time-frequency resource. In the *i. i. d.* case, the more UEs served by the BS, the higher the SE. No limitations when enough many antennas equipped at the BS.
- **Multi-point to multi-point MIMO (Cell-Free) **:** 😊
Multi-BSs together with MU-MIMO: the SE of the mobile system will be increased linearly with the density of BSs in the *i. i. d.* case.

多用户MIMO/SDMA



无线信道的第二类特征

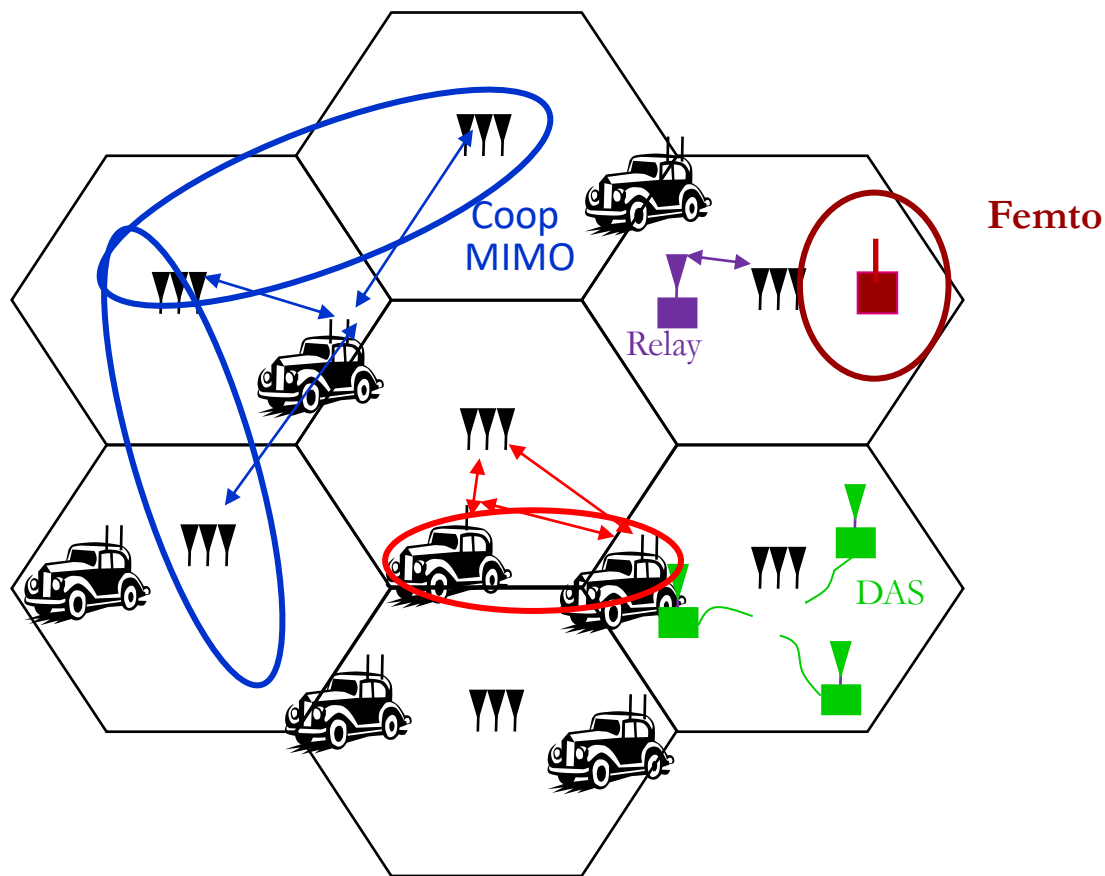
——多用户干扰

- CDMA系统多用户干扰
对抗措施: 多用户检测(MUD)等
- OFDMA系统子载波间干扰
对抗措施: 同步技术、抗频偏技术、干扰删除技术等
- OFDMA系统小区间干扰
对抗措施: 小区间干扰协调(ICIC)、协作多点传输(CoMP)等
- 异构蜂窝网跨层干扰
对抗措施: ICIC、CoMP等
- 大规模MIMO系统导频污染

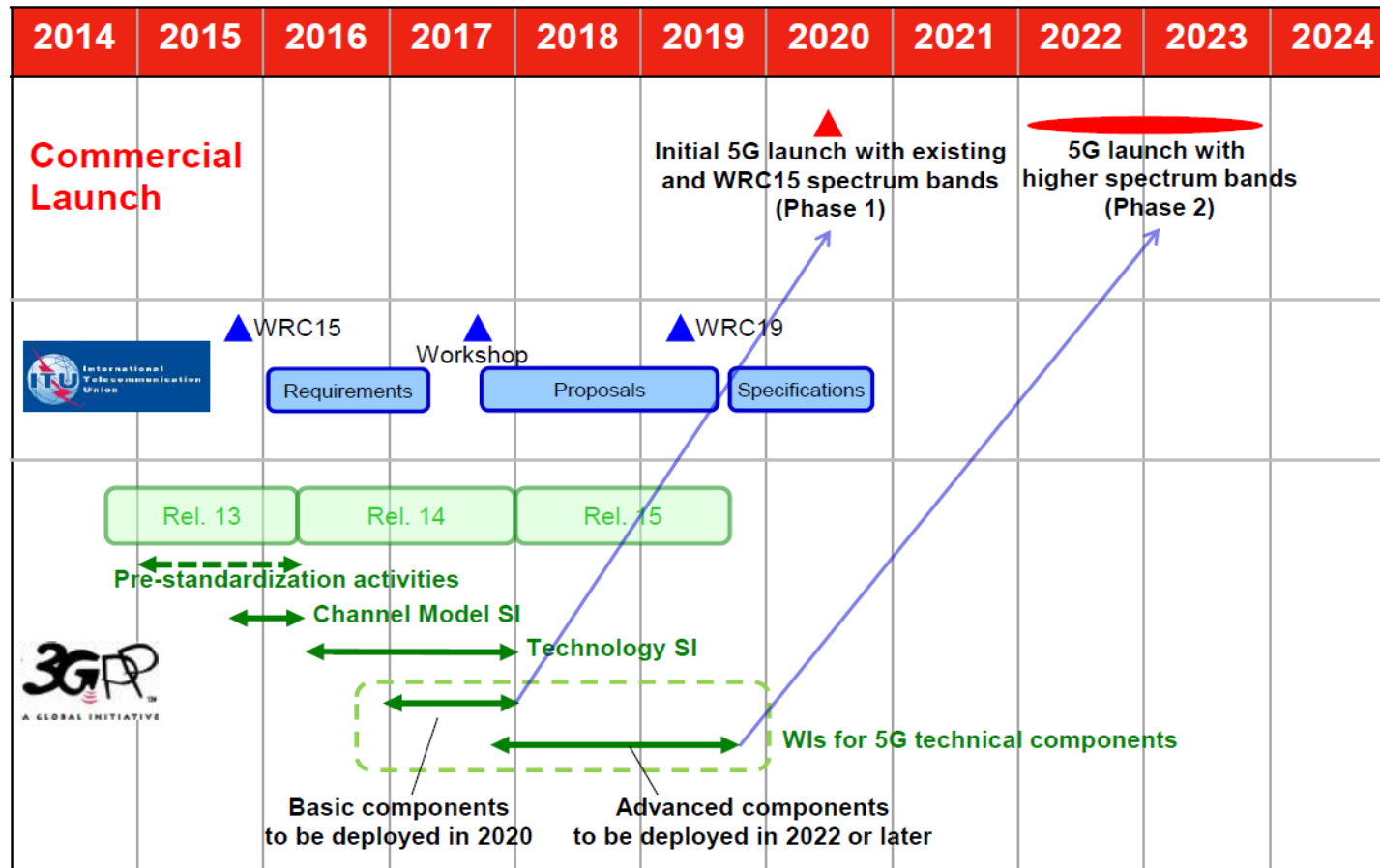
4G 的关键技术

- OFDMA
- MIMO
- 链路自适应技术
- ICIC/CoMP
- 中继
- HetNet

蜂窝新形式



5G 标准化时间表



时代的变迁

Transforming
our world

through intelligent
connected platforms



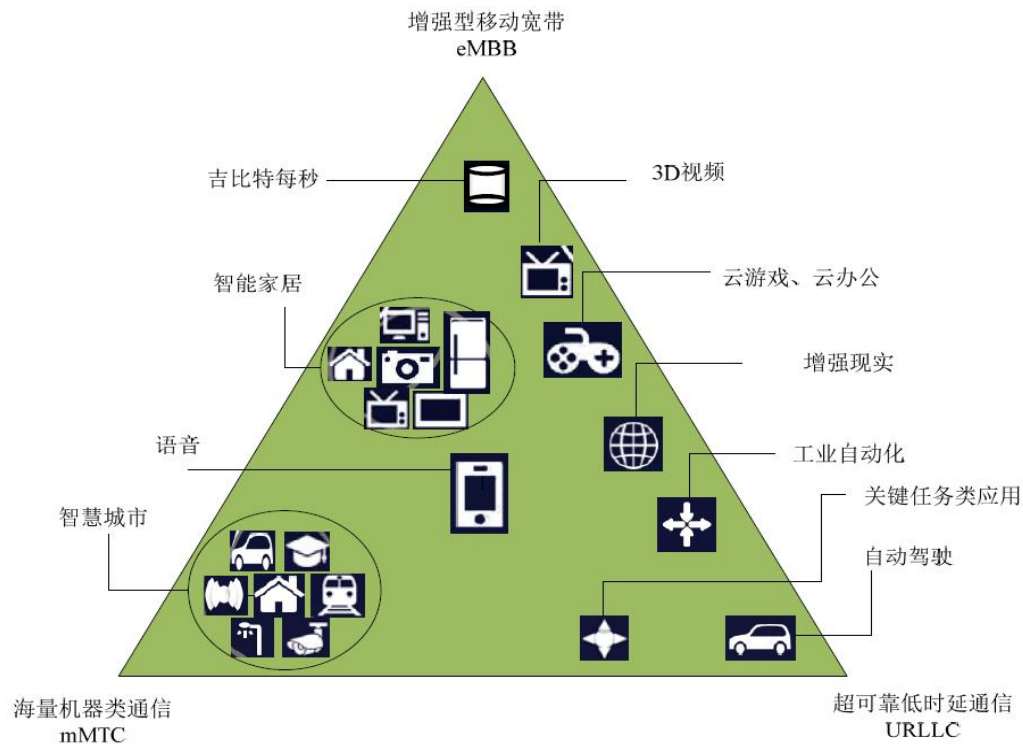
Last 30 years
Interconnecting people



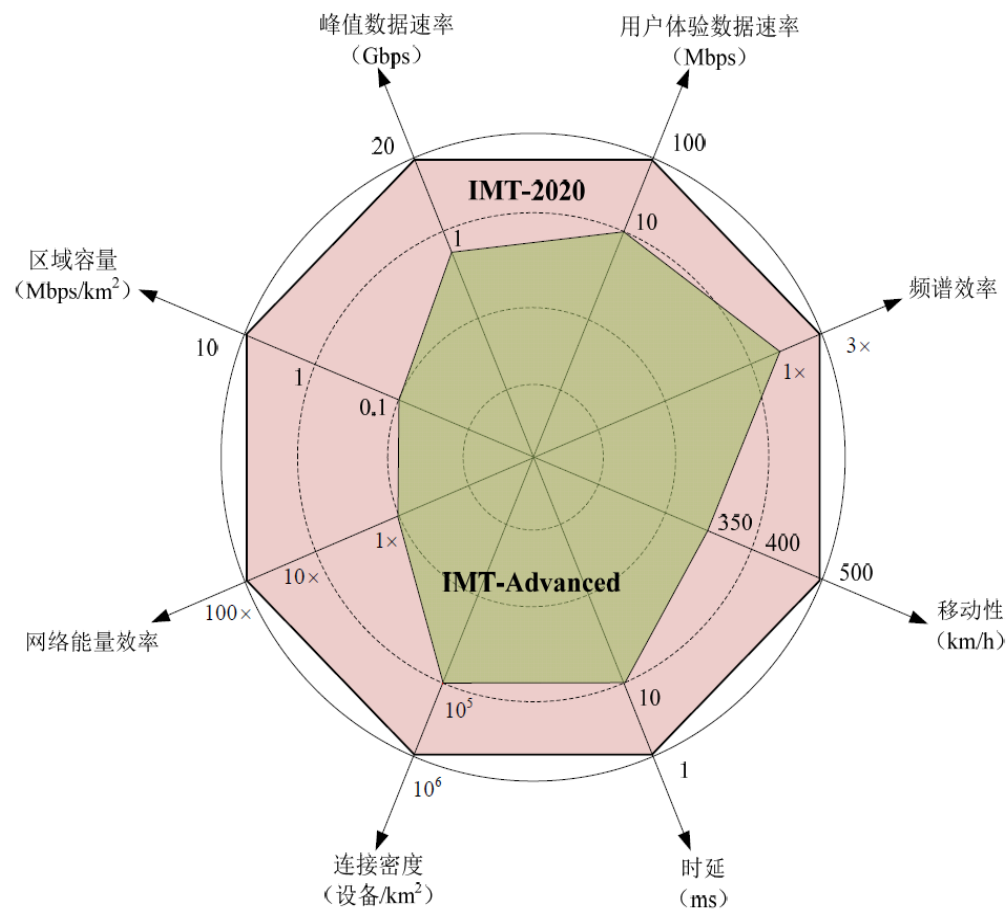
Next 30 years
Interconnecting their worlds

5G 的三个场景

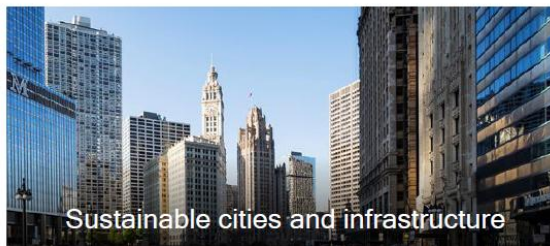
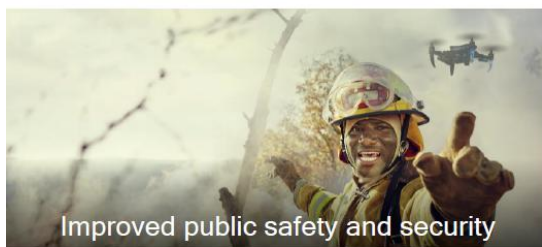
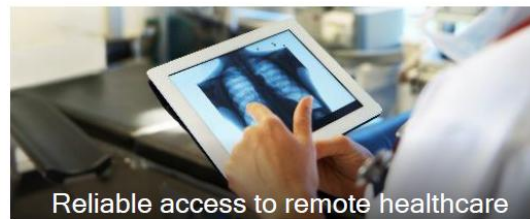
(eMBB, mMTC, URLLC)



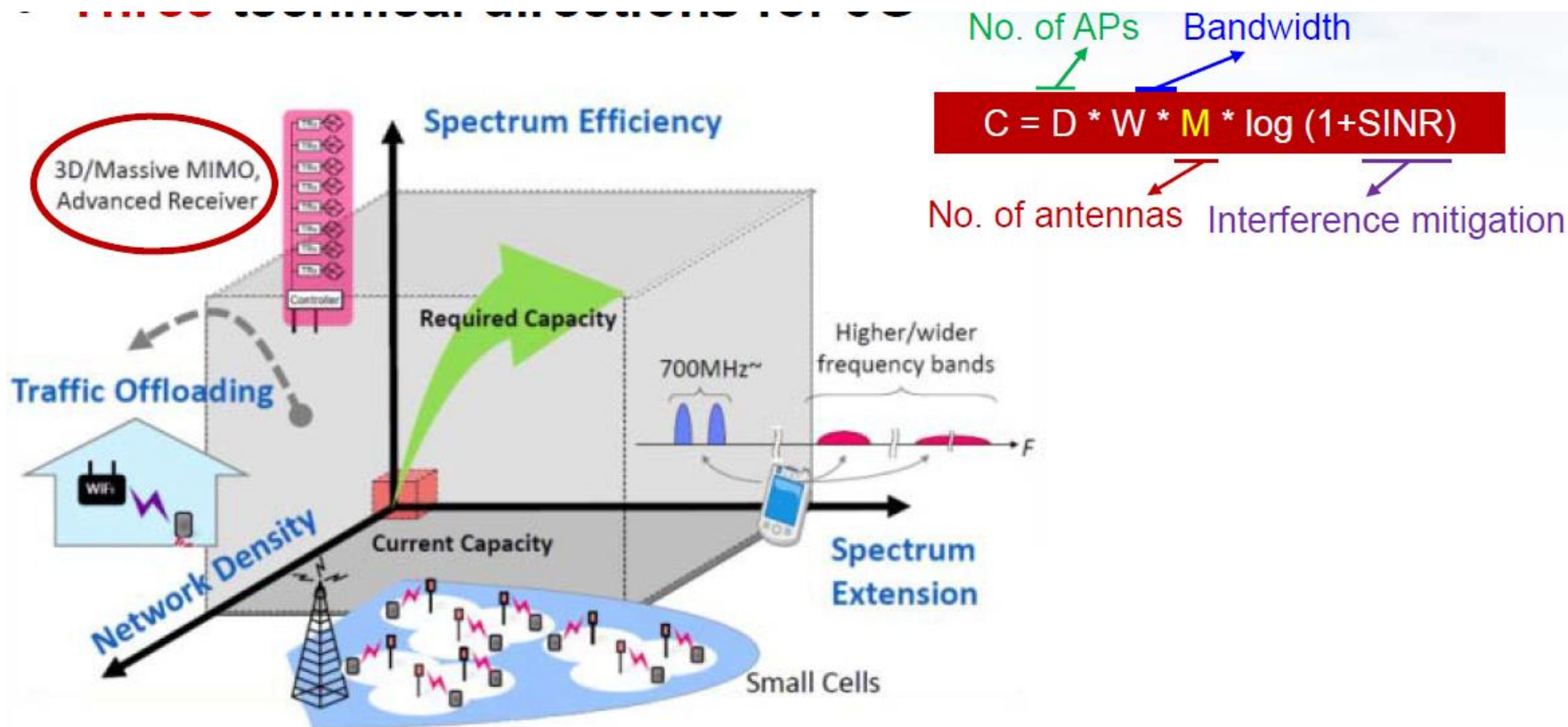
5G 的技术指标



5G 应用



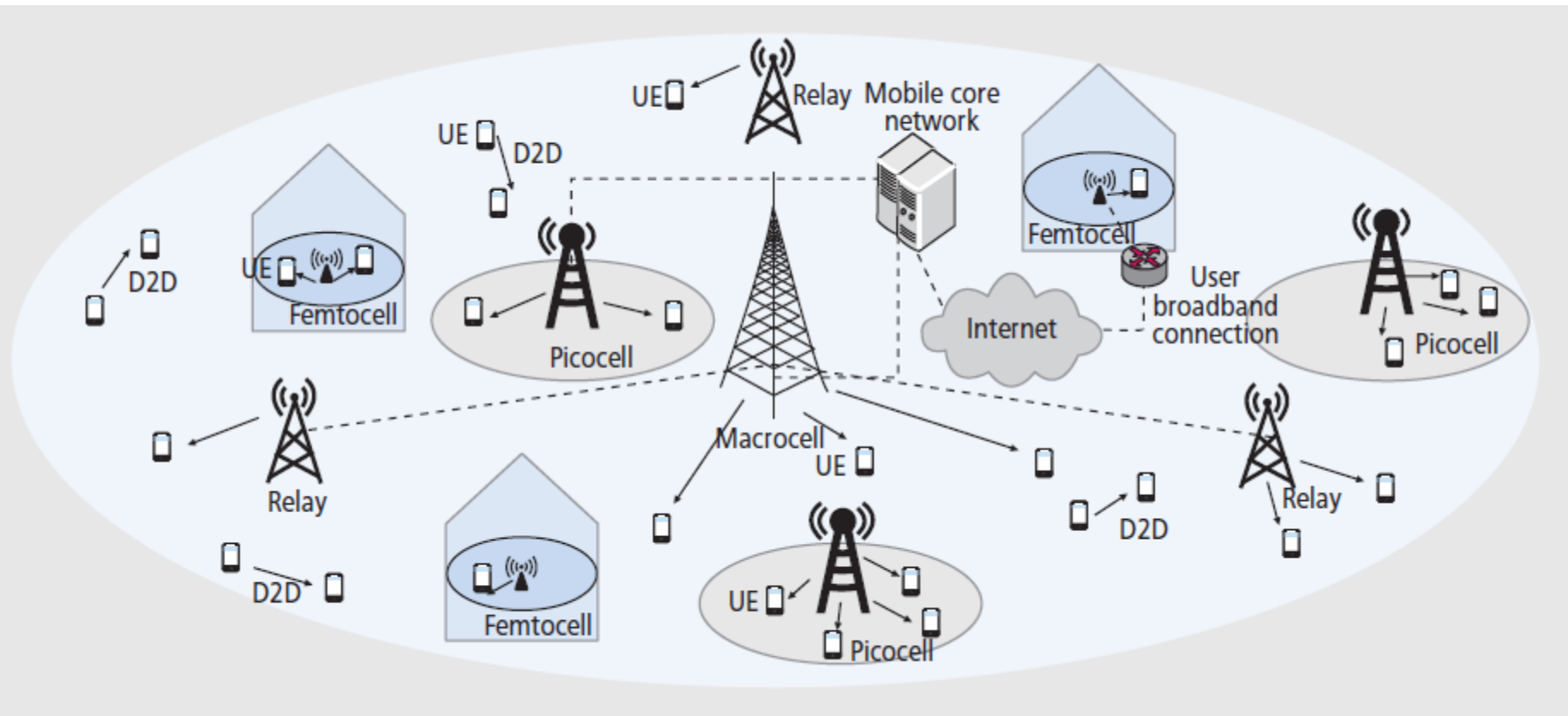
如何实现5G的容量目标？



5G关键技术

- 异构蜂窝网(HetNet)、超密集网(DenseNet)
- 大规模MIMO
- 毫米波通信
- 非正交多址 (NOMA)、非正交波形
- 边缘缓存(Edge Caching)与边缘计算(Edge Computing)

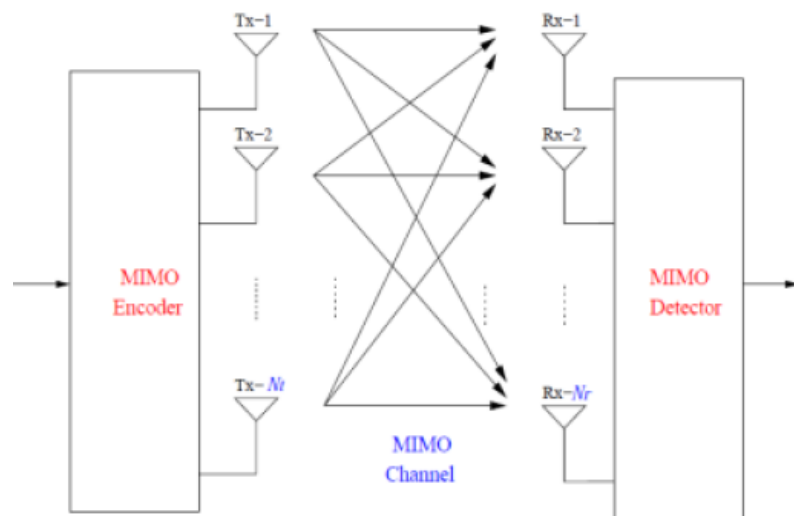
多层HetNet



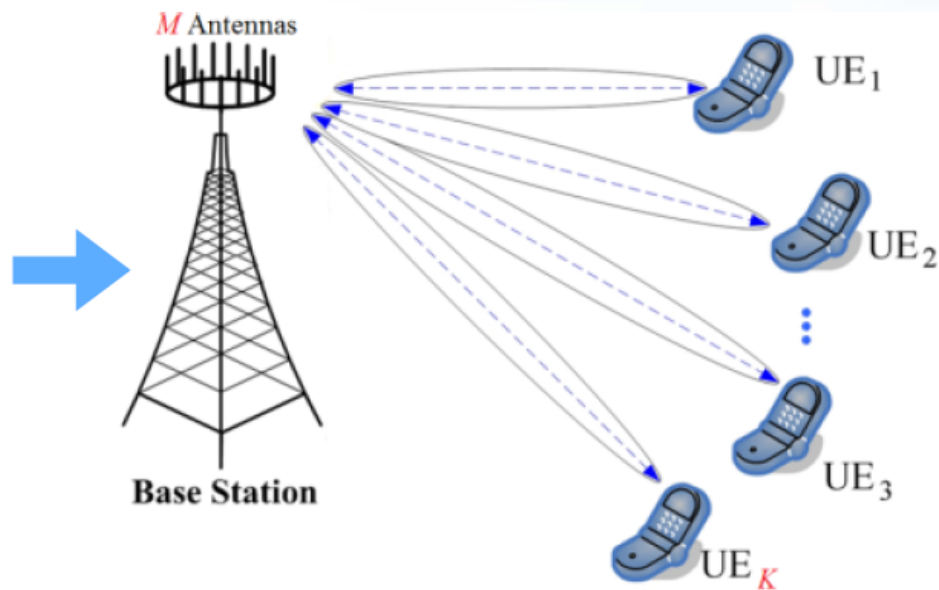
A multi-tier network composed of macrocells, picocells, femtocells, relays, and D2D links.

大规模 MIMO

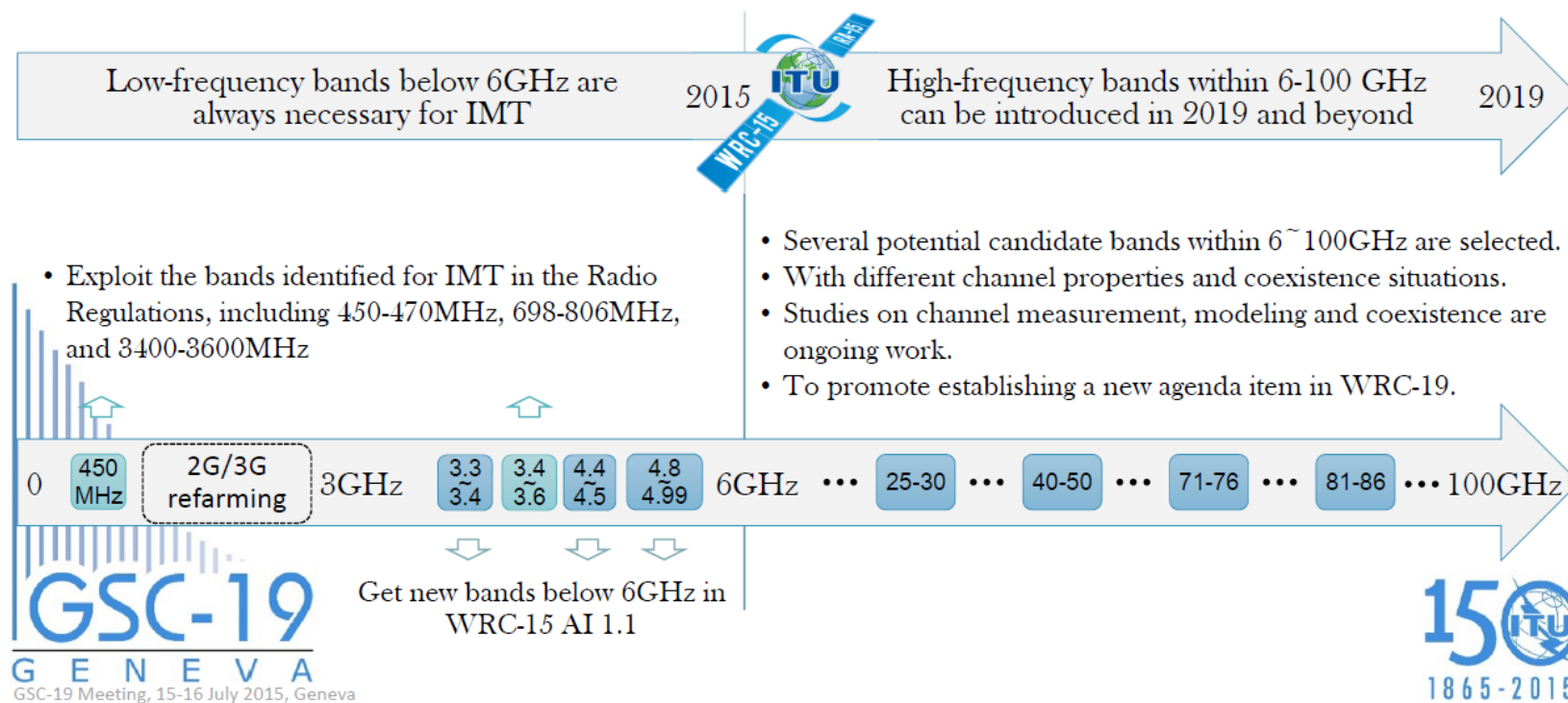
Conventional MIMO
M: 2~8, K: 1~4 (LTE-A)



Massive MIMO
M: ~100~1000, K: 16~64



5G潜在频段



NOMA 标准化

**Study Item
(SI)**



**Technical
Report**



**Work Item
(WI)**



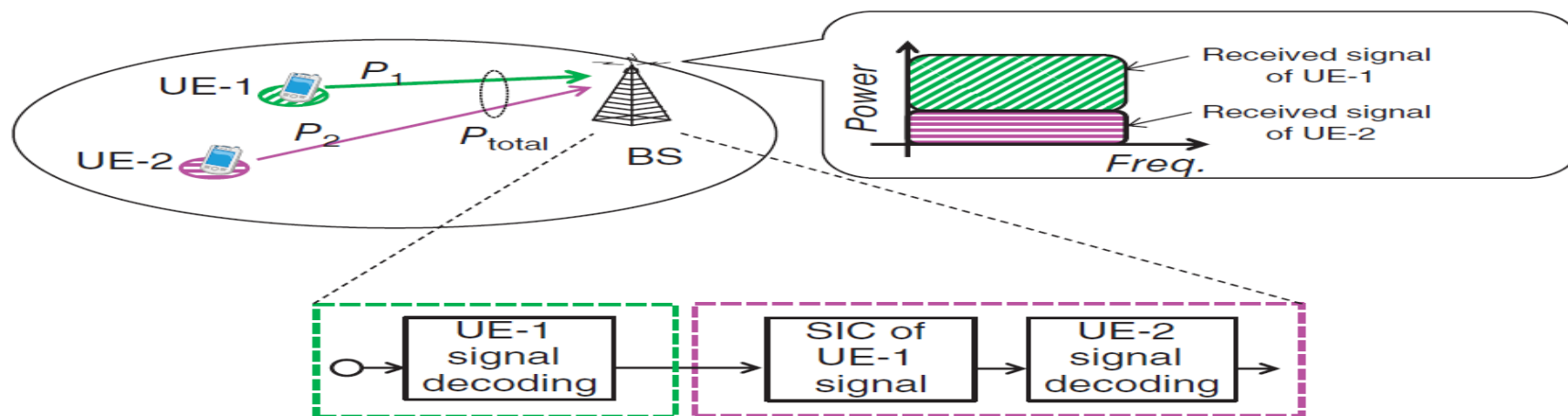
**Technical
Specification**

5G has not yet adopted NOMA completely;

3GPP LTE-A Release-13 SI: Downlink NOMA (MUST);

3GPP NR Release-14 WI: Downlink NOMA (MUST).

上行功率域 NOMA

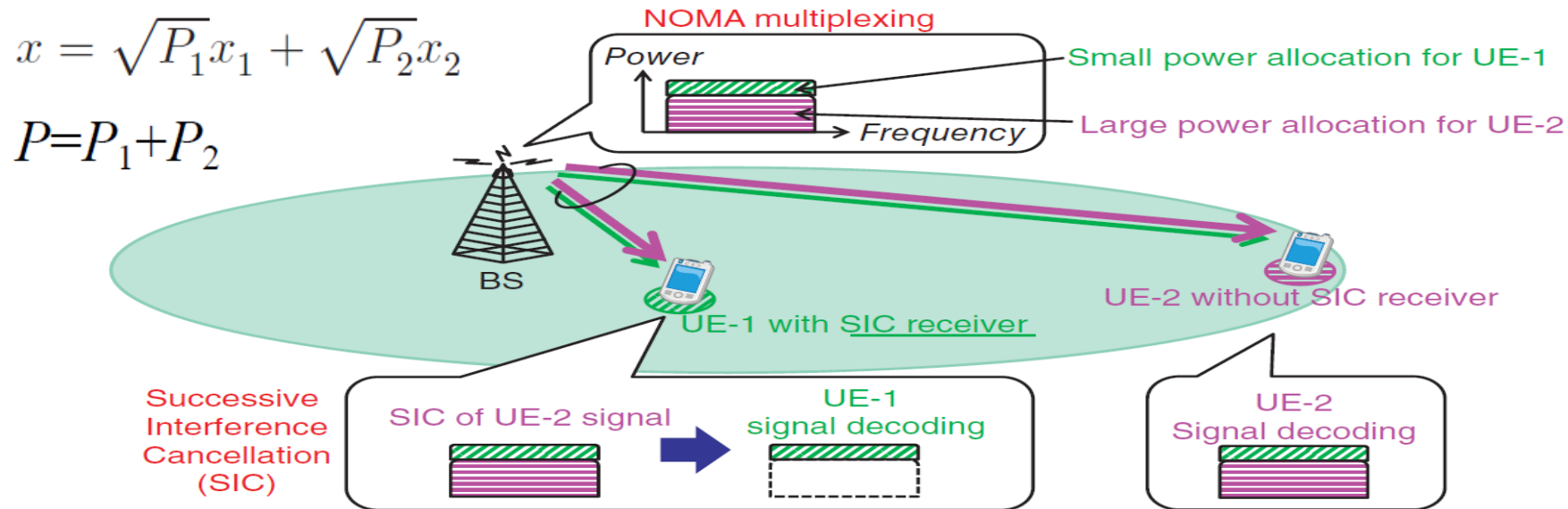


$$R_1 = \log_2 \left(1 + \frac{P_1 |h_1|^2}{P_2 |h_2|^2 + N_0} \right), \quad R_2 = \log_2 \left(1 + \frac{P_2 |h_2|^2}{N_0} \right) \quad (\text{对应 A 点})$$

$$R_1 = \log_2 \left(1 + \frac{P_1 |h_1|^2}{N_0} \right), \quad R_2 = \log_2 \left(1 + \frac{P_2 |h_2|^2}{P_1 |h_1|^2 + N_0} \right) \quad (\text{对应 B 点})$$

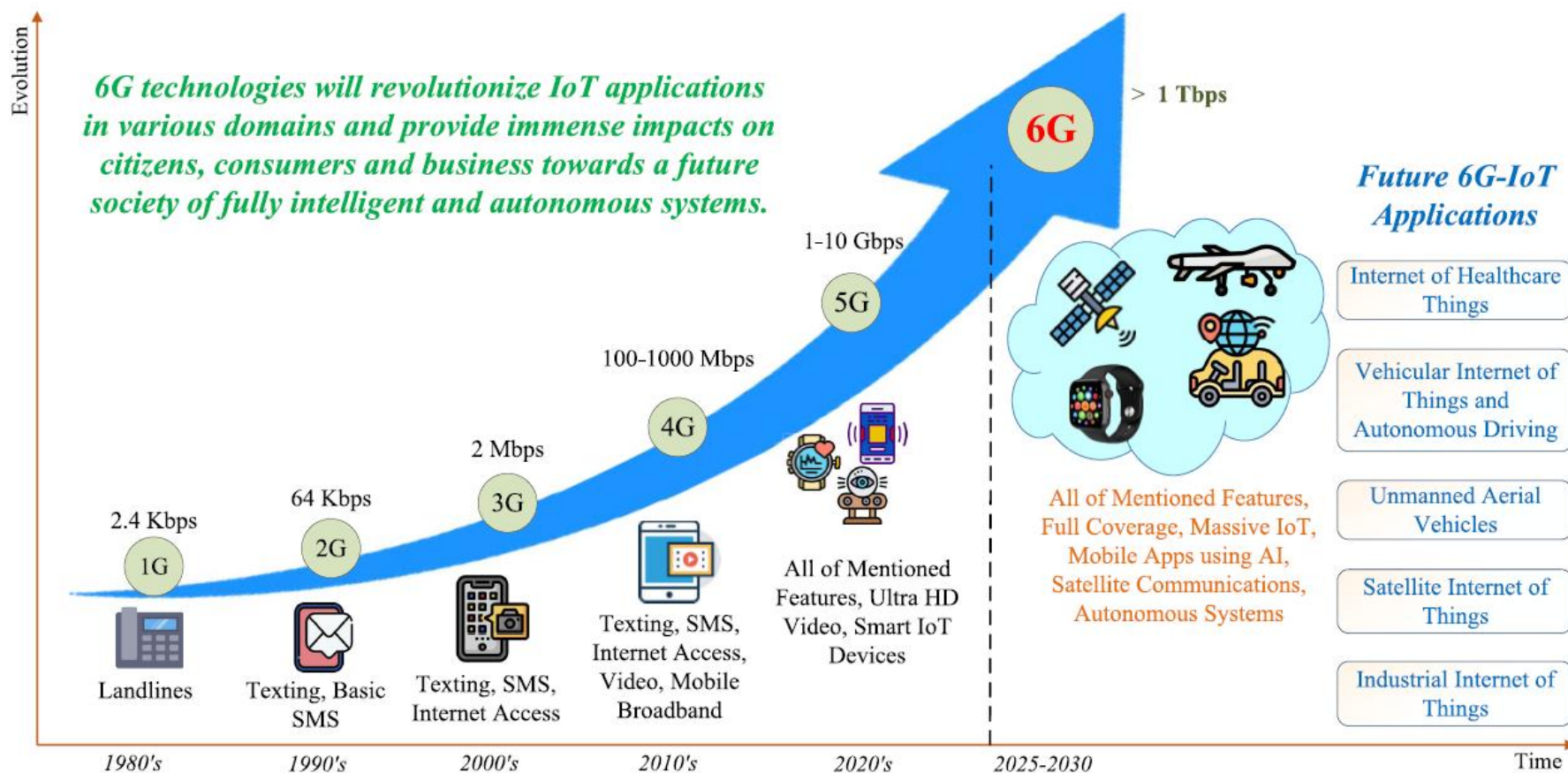
$$R_1 + R_2 = \log_2 \left(1 + \frac{P_1 |h_1|^2 + P_2 |h_2|^2}{N_0} \right)$$

下行功率域 NOMA



$$R_1 = \log_2 \left(1 + \frac{P_1 |h_1|^2}{N_{0,1}} \right), R_2 = \log_2 \left(1 + \frac{P_2 |h_2|^2}{P_1 |h_2|^2 + N_{0,2}} \right)$$

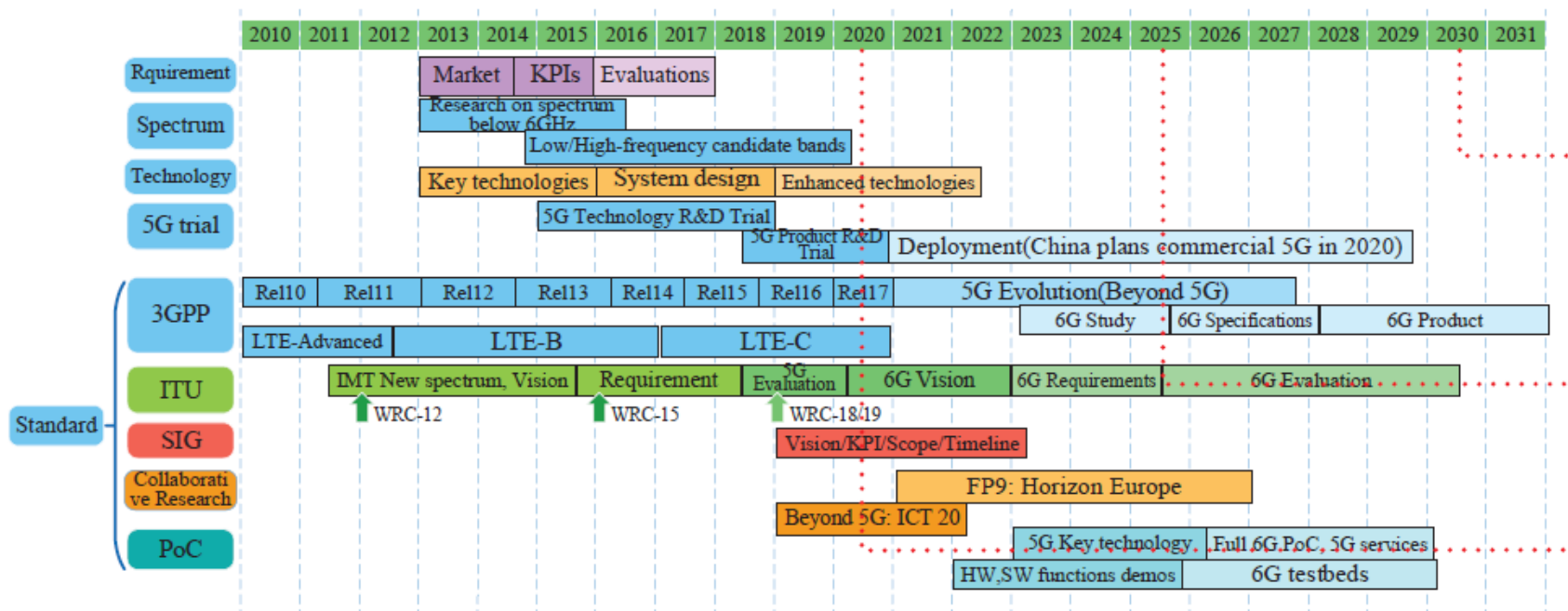
向6G演进



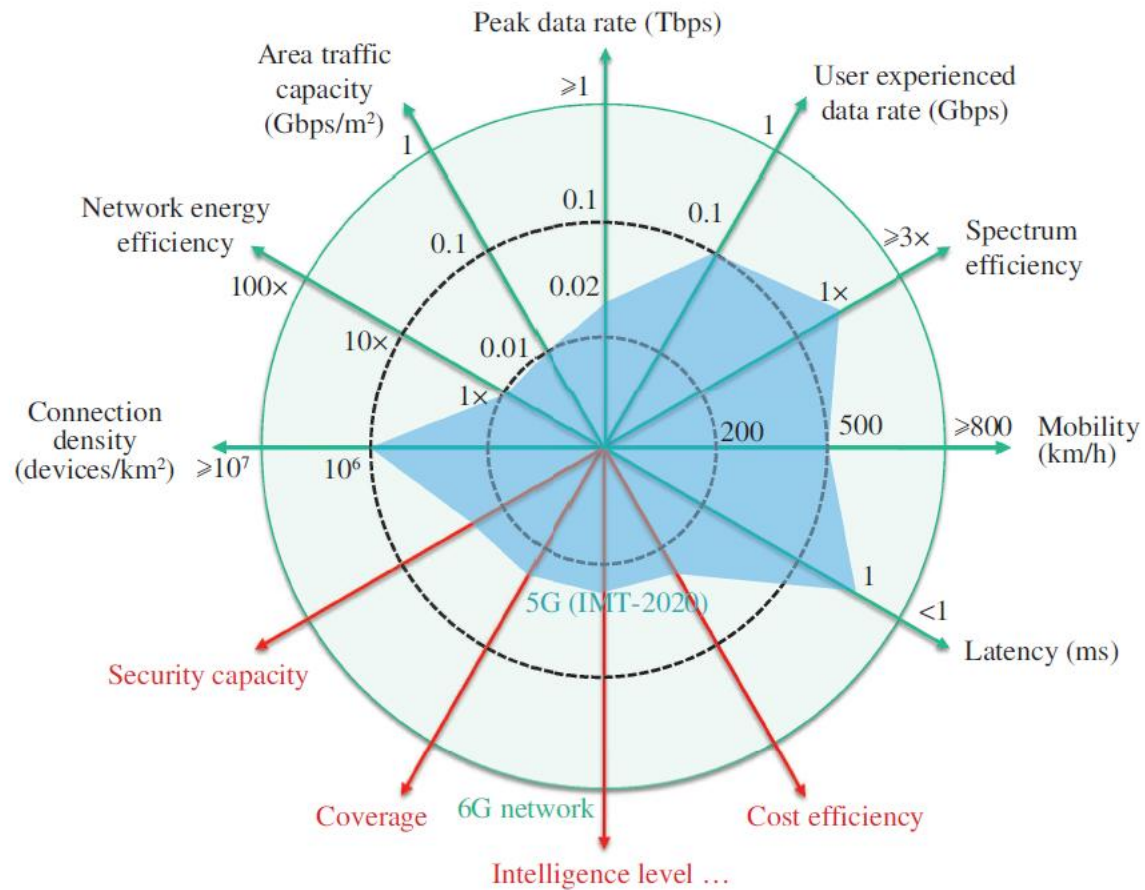
从1G 到 6G

Features	1G	2G	3G	4G	5G	6G (supposed)
Period	1980-1990	1990-2000	2000-2010	2010-2020	2020-2030	2030-2040
Maximum rate	2.4 Kbps	144 Kbps	2 Mbps	1 Gbps	35.46 Gbps	100 Gbps
Maximum frequency	894 MHz	1900 MHz	2100 MHz	6 GHz	90 GHz	10 THz
Service level	Voice	Text	Picture	Video	3D VR/AR	Tactile
Standards	MTS, AMPS, IMTS, PTT	GSM, IS-95, CDMA, EDGE	UMTS, WCDMA, IMT2000, CDMA2000, TD-SCDMA	WiMAX, LTE, LTE-A	5G NR, WWW	—
Multiplexing	FDMA	FDMA, TDMA	CDMA	OFDMA	OFDMA	Smart OFDMA Plus IM
Architecture	SISO	SISO	SISO	MIMO	Massive MIMO	Intelligent surface
Core network	PSTN	PSTN	Packet N/W	Internet	Internet, IoT	IoE
Highlight	Mobility	Digitization	Internet	Real-time streaming	Extremely high rate	Security, secrecy, privacy

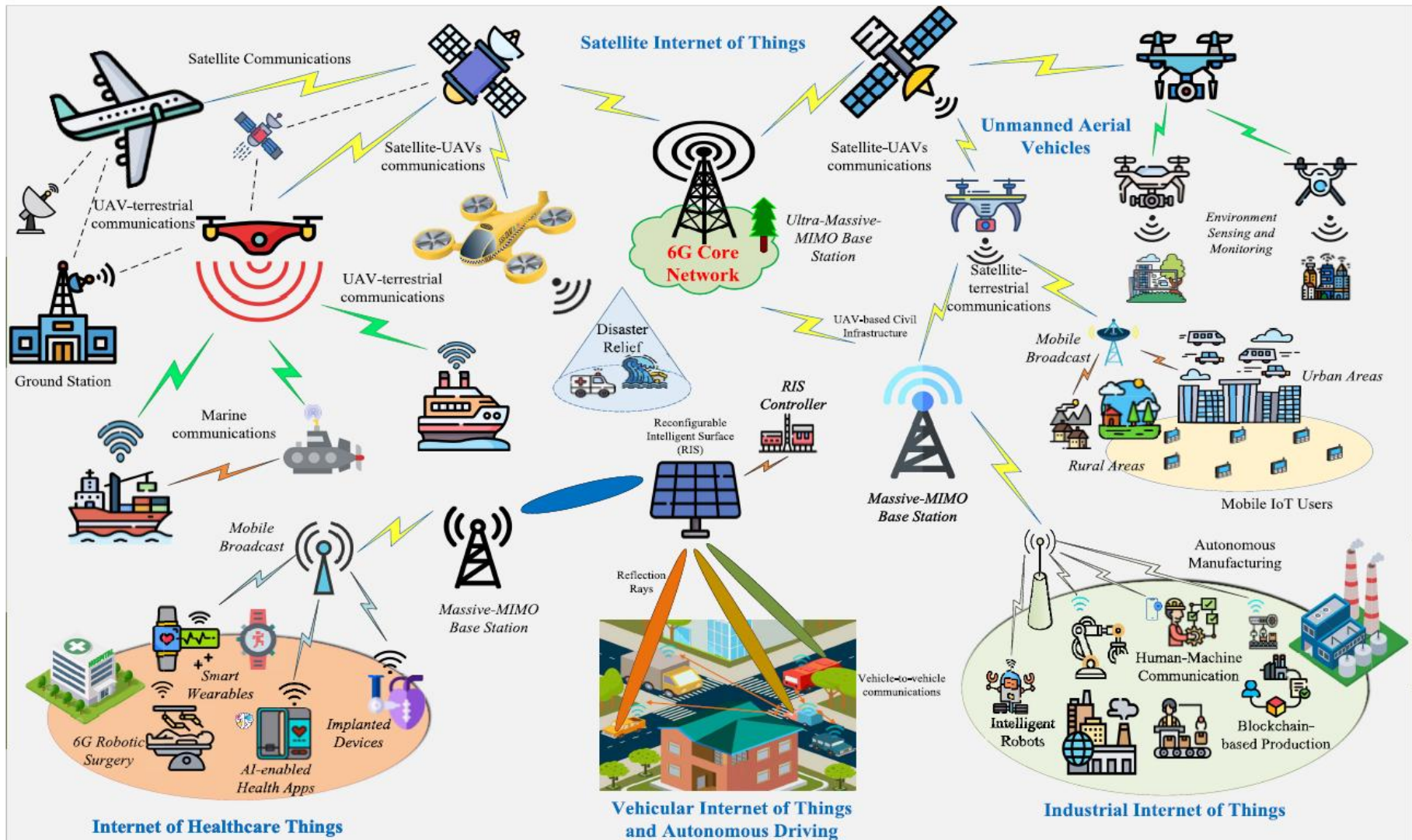
6G 时间表



6G 的技术指标



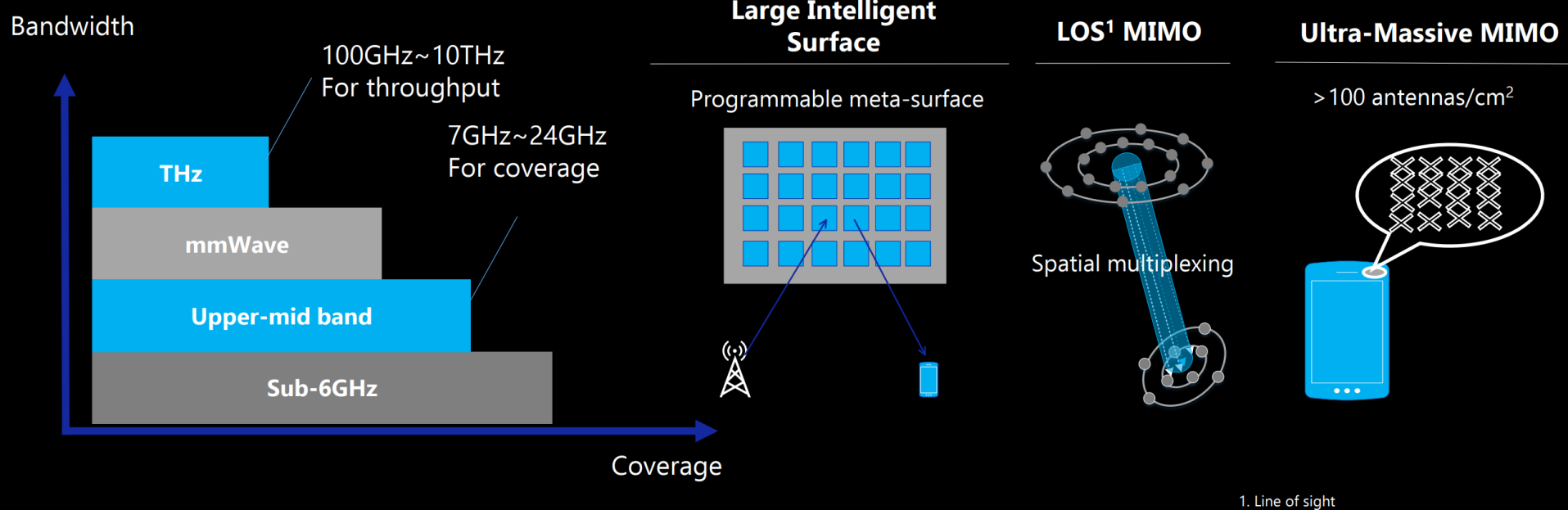
6G 应用愿景



6G 关键技术

More Spectrum and More Antenna

- THz provides large bandwidth, but challenges are there for RFIC/transceiver design and coverage
- Upper-mid band will provide an anchor for coverage
- Large amount of advanced antennas is critical for reliable communication in such 6G spectrum

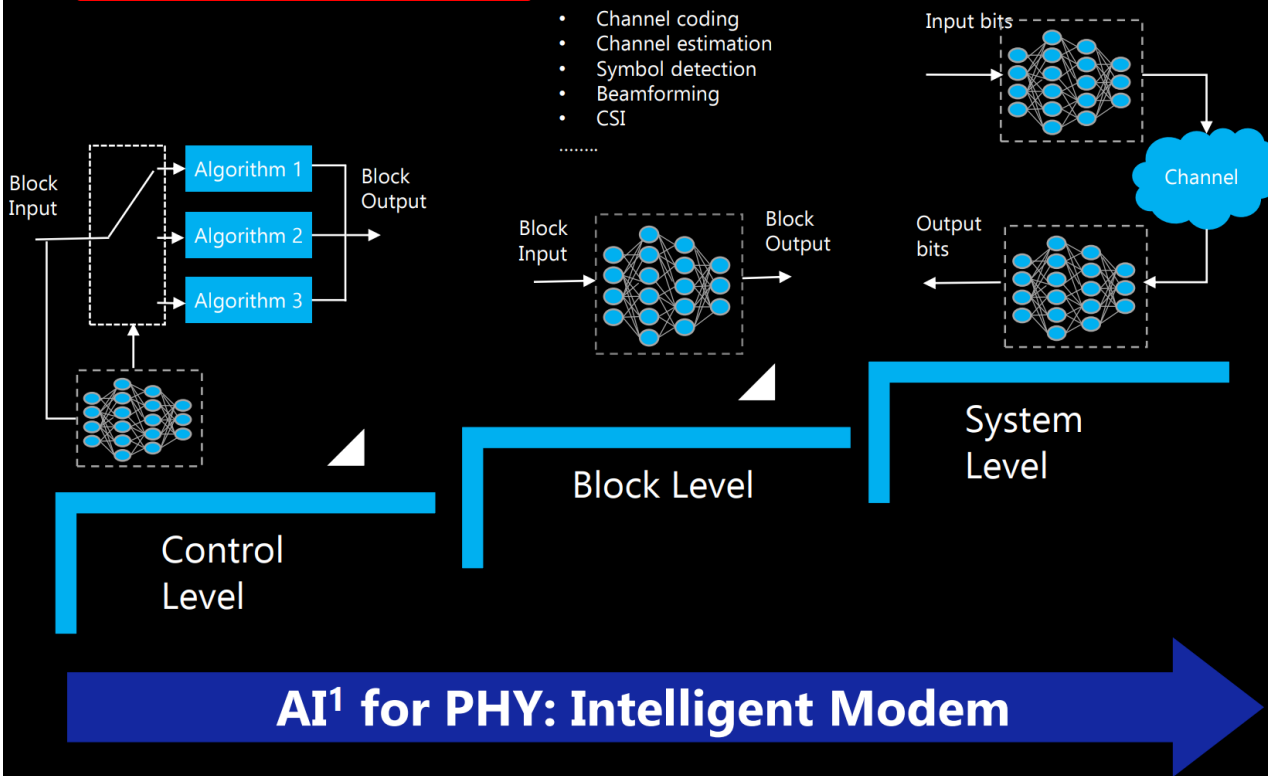


设计新波形（调制）的动机

- OFDM shortcomings:
 - (i) Susceptible to timing and frequency synchronization errors, unsuitable: machine type communications, IoT, high speed com
 - (ii) Large peak to average power ratio, unsuitable: D2D
 - (iii) Large out-of-band radiation, unsuitable: carrier aggregation, opportunistic spectrum access
- OFDM seems not to be suitable to meet the application requirements of 6G.
- Non-orthogonal waveforms (FBMC, GFDM) and orthogonal time frequency space (OTFS) modulation are needed.

6G 关键技术

Higher Intelligence



- **AI for PHY in 5G advanced**
 - CSI
 - Beam management
 - Positioning
- **AI native PHY for 6G**
- **Comprehensive AI in all layers**
 - End-to-End AI
 - Split computing
 - Federated learning

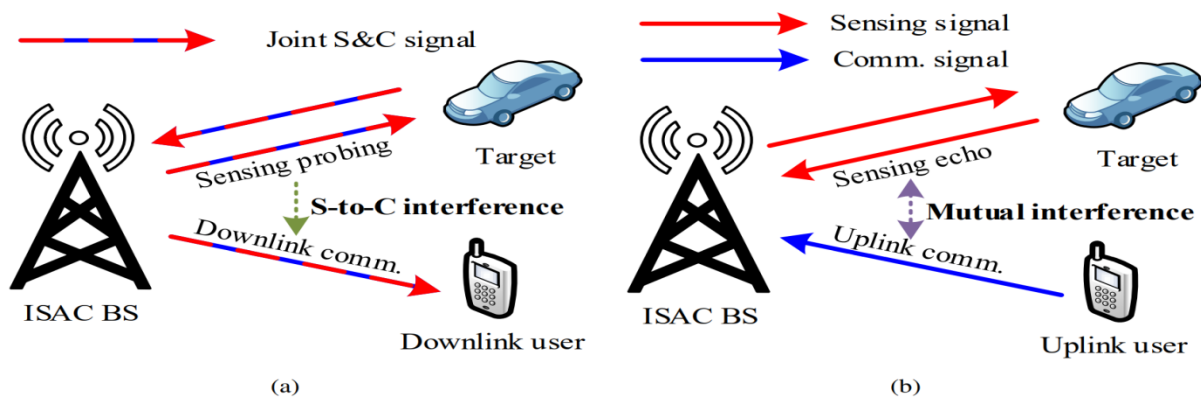
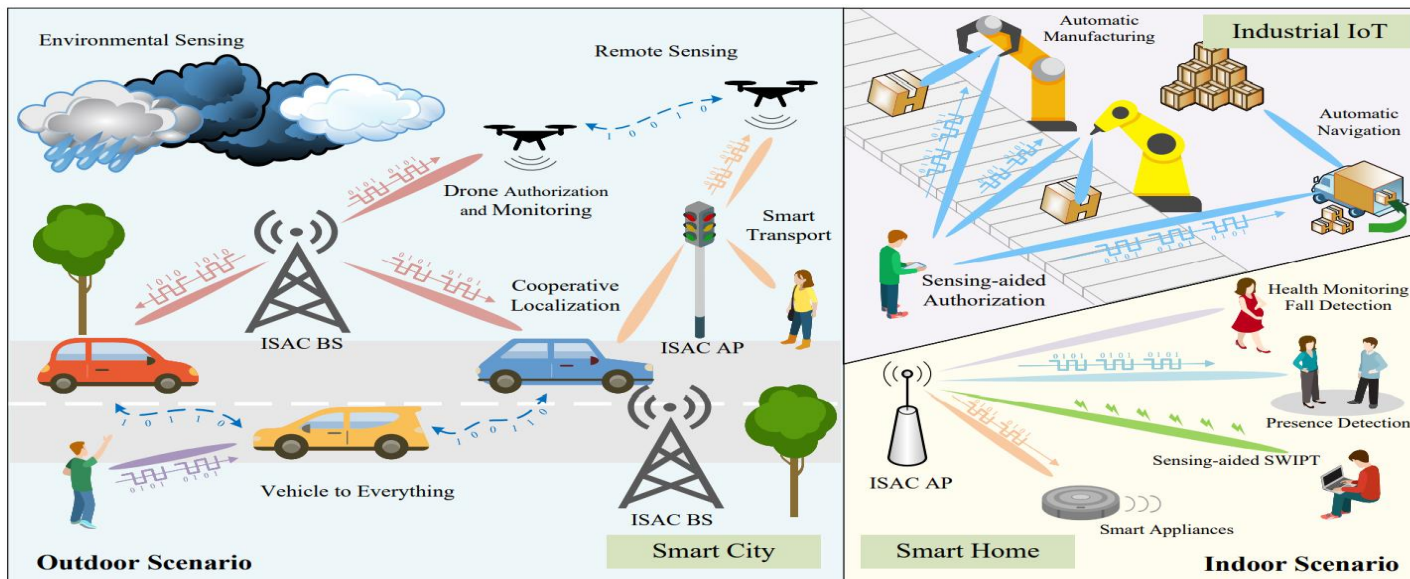
6G 关键技术

- 空天地海(Space-Air-Ground-Sea)通信一体化架构
- 感知通信一体化(ISAC)架构
- 无蜂窝(Cell-Free)网络架构

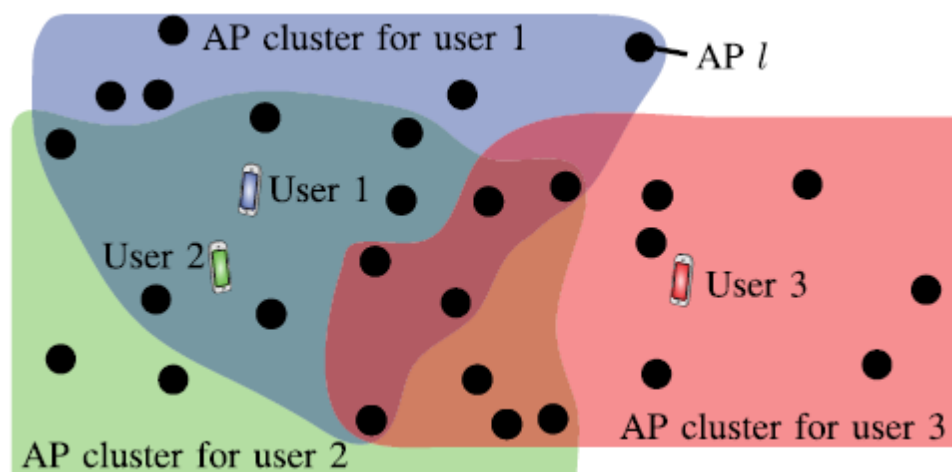
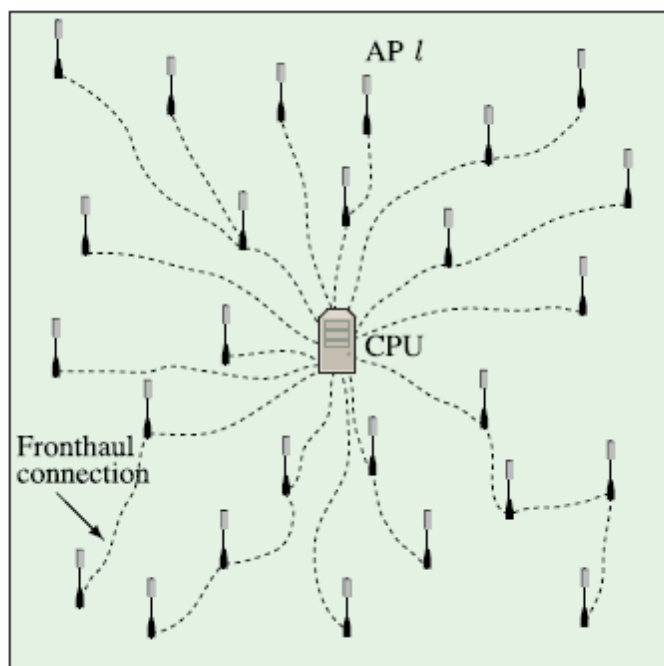
空天地海一体化



感知通信一体化



无蜂窝网络架构



Thanks